paceval.

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Reusable Mathematical Execution for Collider Research

An open research proposal from paceval UG for experimental particle physics, including CERN collaborations and research institutions working with modern particle accelerators.

> Could reusable mathematical execution graphs help collider researchers evaluate large event samples more efficiently and assess numerically sensitive results with greater confidence?

Modern collider experiments extract physical knowledge from large numbers of reconstructed events. Analysis workflows repeatedly calculate kinematic observables, correction functions and parametrized models. These operations already benefit from highly optimized software. Nevertheless, a focused investigation could establish whether an additional mathematical execution engine provides measurable scientific value for carefully selected tasks. [1, 2, 3, 4, 5]

paceval is a mathematical runtime engine developed by paceval UG. It turns explicit mathematical expressions into reusable execution graphs that can be evaluated again with new input values, including batches of values. Its documentation also describes Trusted Interval Computing (TINC), a method intended to provide numerical interval information. [6, 7, 8]

We propose an open and reproducible study alongside established collider software. The starting point is deliberately focused: scalar computation kernels relevant to event analysis, openly available CMS data and independently constructed numerical stress tests. If the results are promising, the collaboration could examine more complex functions, calibration models and hardware execution.

Research at a glance

- Computational performance. Compare selected expressions evaluated by paceval with optimized C++ and ROOT VecOps.
- Numerical quality. Assess mathematically equivalent formulations and examine TINC results against independent high-precision and interval references.
- Scientific applicability. Determine whether a validated component can be integrated without changing established event reconstruction or analysis methods.

This proposal is intended for research groups. paceval UG is willing to make its technology available for an agreed scientific pilot study. We also welcome joint development of interfaces and additional mathematical functionality when this serves a concrete research need.

1 Scientific context and objective

The High-Luminosity Large Hadron Collider (HL-LHC) is planned to support much larger datasets from around 2030 onward. The CMS Offline Software and Computing Conceptual Design Report, published in March 2026, identifies increasing detector complexity, data volume and computing demands. It describes a technical roadmap for preparing the computing ecosystem. [1, 2]

The LHCb Allen project demonstrates the importance of graphics processing units for real-time event selection. ROOT provides a mature environment for collider data analysis, including multithreading, just-in-time compilation of C++ expressions and vector-oriented operations. Our proposal does not seek to replace these systems. It explores whether a separate mathematical kernel can deliver additional throughput, better numerical diagnostics or simpler reuse for defined tasks within existing workflows. [3, 4, 5]

What capabilities does paceval provide?

The published paceval application programming interface (API) describes the creation of computation objects from formulas and their reuse for different inputs. It also supports repeated and batch evaluation. For potential collider integration, the C and C++ interfaces and the ability to reuse a formula across large input sets are especially relevant. [6, 7]

Trusted Interval Computing (TINC) is the name used by paceval for its numerical interval method. An important research task is to determine under which conditions its intervals rigorously enclose the intended exact result. An independent reference is the interval or ball arithmetic implemented by FLINT/Arb, for which the inclusion principle is documented. [8, 9]

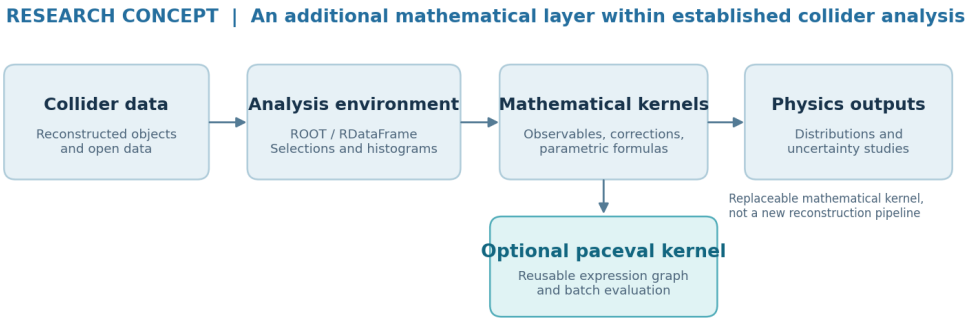


Figure 1. Conceptual integration of paceval into an existing collider analysis. Established data formats, selections, histograms and physics models remain in place. paceval would be evaluated as an optional mathematical execution component.

The opportunity for particle physics research

A mathematical kernel that proves faster or provides stronger numerical diagnostics on specific collider workloads could reduce analysis costs, support broader parameter studies and help investigate numerically sensitive results. Because paceval represents mathematical expressions as reusable execution graphs, selected kernels could also be examined for future implementation on field-programmable gate arrays (FPGAs) or application-specific integrated circuits (ASICs). Such hardware may offer higher throughput, lower latency or improved energy efficiency if data movement and numerical requirements are appropriately addressed. These are measurable research objectives rather than assumed performance gains.

2 Application areas with clear research questions

Application area	Potential paceval contribution	Potential scientific value
Kinematic observables	Repeated evaluation of formulas for invariant masses, angles and event selections.	Higher throughput or better-documented numerical edge cases.
Corrections and event weights	Reuse analytic functions across events and parameter sets.	Efficient systematic studies with reproducible expressions.
Numerical validation	Independently evaluate sensitive formulas and TINC interval results.	Greater insight into rounding-related effects.
Hardware-oriented kernels	Assess selected graphs for FPGA execution after software validation.	Longer-term option for predictable computational kernels.

Not every collider task is a suitable first target. Pattern recognition, full track reconstruction, large matrix computations and detailed detector simulation rely on specialized methods. An initial evaluation of paceval is most informative when the calculation is scalar, repeated frequently and defined by a mathematical expression with an independent reference. [3, 4, 5, 10]

3 Work package A: Kinematic observables and numerical stability

For two particles whose masses can be neglected, the squared invariant mass can be calculated from transverse momenta and angular variables. In natural units, where c = 1, and with Δη denoting the pseudorapidity difference and Δφ the azimuthal angle difference, the expression is:

$$m^2 = 2\, p_{t1}\, p_{t2}\, [\cosh(\Delta\eta) - \cos(\Delta\varphi)]$$

The following expression is algebraically equivalent:

$$m^2 = 4\, p_{t1}\, p_{t2}\, [\sinh^2(\Delta\eta/2) + \sin^2(\Delta\varphi/2)]$$

Both equations apply under the same massless-particle approximation. For nearly collinear momenta, the first form can lose substantial relative precision because it subtracts two values that are both close to one. The second form avoids that particular cancellation. This established numerical principle provides a well-defined and physics-relevant way to test whether paceval can offer efficient evaluation together with useful numerical diagnostics.

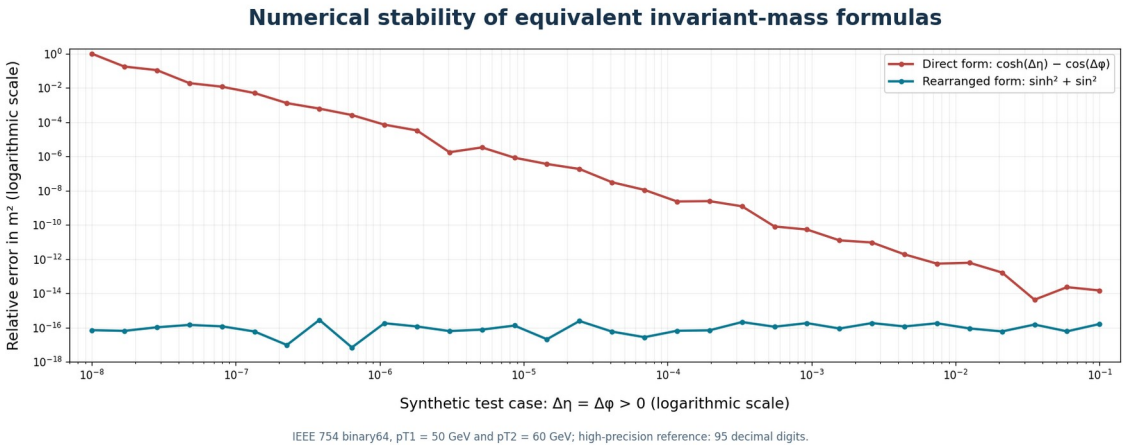


Figure 2. Independently calculated illustrative IEEE 754 binary64 example with pT1 = 50 GeV, pT2 = 60 GeV and Δη = Δφ. The numerical reference uses 95 decimal digits. The plot shows a numerical test, not a paceval benchmark or an experimental CERN measurement.

The proposed study would assess how paceval evaluates both formulations and whether TINC supplies useful diagnostic information for sensitive inputs. Test cases would include near-collinear configurations, selection thresholds and representative collision events. We would also examine whether numerical differences affect event selections or distributions in a physically meaningful way.

4 Work package B: Repeated evaluation of parametric functions

Collider analyses use calibration corrections, efficiency factors, event weights and systematic variations. ROOT RDataFrame already supports variations of input columns within a shared analysis workflow. A meaningful pilot study should therefore focus on repeatedly evaluated, analytically defined functions and compare them with the capabilities of existing tools. [4, 11]

Testing can begin with an explicitly synthetic function that is not claimed to represent a calibrated detector. A research partner could then supply an approved function with demonstrable physical relevance. For both cases, the study would separately time initialization, mathematical evaluation, data conversion and end-to-end processing.

A useful research outcome would be a validated expression that can be reused throughout a parameter scan without regenerating the mathematical graph. Its practical value must be measured against an optimized C++ or ROOT implementation that computes the same quantity with comparable numerical requirements.

5 Work package C: Numerical quality and reproducibility

Statistical uncertainty, detector calibration uncertainty, physics model uncertainty and numerical rounding error are different concepts. A mathematical graph cannot remove physical uncertainty. It can, however, provide a controlled way to study the numerical behavior of a classical computation.

We propose to compare paceval results against high-precision reference values from MPFR and independently computed rigorous enclosures from FLINT/Arb. The study would record absolute and relative error, error measured in units in the last place (ULP), exceptional cases and interval width. TINC output would be assessed rather than treated as an independently certified error guarantee. [8, 9, 13]

We would also test reproducibility under documented compiler settings, rounding modes, degrees of parallelism and processor architectures. This would establish the conditions under which results are repeatable and clarify any numerical differences between platforms. Cross-platform bitwise identity is not assumed.

6 Technical pilot study and fair comparison

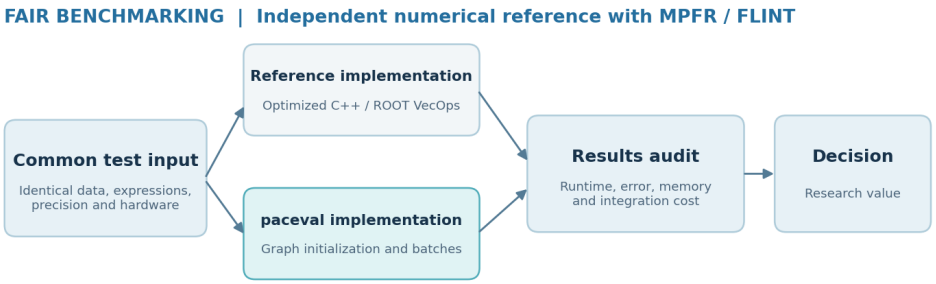


Figure 3. Proposed benchmark design using identical input data. Numerical correctness, execution time and data movement are evaluated separately. The software comparison includes both initial setup and repeated batch evaluation.

Data and software environment

Public CMS NanoAOD datasets offer a practical starting point. The CERN Open Data Portal explains how NanoAOD data can be processed with ROOT and compatible Python libraries. A reproducible sample can be constructed with documented dataset versions and selection conditions. Synthetic numerical edge cases would complement the real data because they may occur only rarely in typical samples. [12]

Comparison path	Role in the pilot study
C++ and ROOT RDataFrame	Production-oriented reference using optimized C++ and ROOT VecOps. [4, 5]
paceval C/C++ API	Initialize reusable graphs, evaluate batches and return the results to the analysis. [6, 7]
MPFR and FLINT/Arb	Independent high-precision values and rigorous interval enclosures. [9, 13]
Optional additional backend	GPU comparison when justified by the selected workload. [3]

Measurement protocol

- Numerical comparability. Use identical expression definitions, units, input values, approximations and documented precision requirements.
- Performance. Measure graph construction, evaluation, data transfer, total runtime, memory usage and scaling with the number of events separately.
- Fair baselines. Use comparable processors and parallelism, optimized compiler settings, and vectorization when appropriate for the reference implementation.
- Physics relevance. Measure changes in selection outcomes, distributions and derived parameter estimates separately from numerical error itself.
- Reproducibility. Publish the relevant source code, build parameters, dataset version and experimental protocol wherever permitted.

The pilot would be scientifically informative if a representative kernel demonstrates a reproducible benefit that matters for the research workflow. This benefit might consist of reduced runtime at comparable accuracy, additional reliable numerical diagnostics or a more convenient way to maintain and reuse a validated expression. Negative results would also delineate the useful limits of the proposed architecture.

7 Relationship to existing collider systems and hardware outlook

ROOT already provides vectorized and multithreaded analysis capabilities. LHCb Allen demonstrates an established GPU-based approach to triggering. Other specialized tools include hls4ml for selected FPGA inference workloads, Geant4 for particle transport simulation and Minuit2 for numerical optimization. paceval is proposed as an optional mathematical computation layer that can be evaluated alongside these proven tools. [3, 4, 5, 10, 14, 15]

A possible second research stage would evaluate the implementation of selected paceval execution graphs on FPGA or ASIC hardware. Functions evaluated repeatedly across many events are natural candidates for exploration. The study would consider parallel throughput, transfer overhead, available arithmetic precision, latency and energy per result. Only a measured hardware implementation could establish an end-to-end advantage.

Potential follow-on studies include more complex correction functions, repeated parameter optimization and selected algorithms in detector calibration or online analysis. Their priority should follow the scientific needs and workloads identified by a research partner.

Open invitation to collaborate with paceval UG

paceval UG would welcome the opportunity to make its mathematical engine available for research in experimental particle physics. We invite groups at CERN, universities and research institutes to identify suitable tasks, independently test the proposed hypotheses and, where useful, co-develop interfaces or mathematical operations for the API.

We suggest a focused pilot study with transparent evaluation metrics, documented reproducibility and methods suitable for scientific publication. Terms relating to software use, licensing, data and publication can be agreed jointly at an early stage.

> Our objective is to contribute a useful mathematical capability to particle physics research. Rather than replacing existing collider frameworks, we want to explore where reusable mathematical execution and numerical quality control could extend the options available to researchers.

Contact: paceval UG, Munich, Germany | info@paceval.com | https://paceval.com

References and technical resources

[1] CMS Collaboration (2026). CMS Offline Software and Computing for HL-LHC Conceptual Design Report. CERN-LHCC-2026-003.

[2] CERN. High-Luminosity LHC. Project background and schedule.

[3] LHCb Collaboration (2020). Allen: A High-Level Trigger on GPUs for LHCb. Computing and Software for Big Science 4, 7. DOI: 10.1007/s41781-020-00039-7.

[4] ROOT Collaboration. ROOT RDataFrame class reference and data analysis functionality.

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[6] paceval UG. Mathematical Engine and public source repository.

[7] paceval UG. Application programming interface and reusable computations.

[8] paceval UG. Trusted Interval Computing (TINC) product documentation.

[9] FLINT developers. Ball arithmetic, rigorous inclusion and error propagation.

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[12] CERN Open Data Portal. Getting Started with CMS NanoAOD Open Data.

[13] GNU MPFR Project. Multiple-precision floating-point library with correctly rounded operations.

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[15] Lorusso, M. (2023). Implementing Machine Learning Inference on FPGAs from Software to Hardware Using hls4ml. CMS-CR-2023-019.

Glossary

HL-LHC means High-Luminosity Large Hadron Collider. The upgrade creates growing demands for detectors, reconstruction and data analysis.

ROOT is a scientific data analysis software framework widely used in high-energy physics. RDataFrame is its high-level interface for declarative and parallel data processing.

GPU means graphics processing unit, which can execute highly parallel computations. FPGA means field-programmable gate array, a reconfigurable digital hardware device. ASIC means application-specific integrated circuit, a chip built for a defined purpose.

TINC stands for Trusted Interval Computing, the interval method described by paceval. ULP means unit in the last place of a floating-point representation.

MPFR is the Multiple Precision Floating-Point Reliable Library. FLINT includes Arb functionality for rigorous interval and ball arithmetic.

Note. This is an independent research proposal from paceval UG. It does not imply any existing partnership with CERN or a collider collaboration. Performance, numerical quality and integration benefits remain to be evaluated experimentally.